

1.Introduction to GIS

Session 1. Introduction to GIS

Learning Objectives

1. What is GIS
2. Importance of GIS
3. Different types of data used in GIS and when to use what
4. What are the common tools
5. Common sources of Data

What is GIS?

There are a number of definitions of GIS. Different groups of people (general public, planners, teachers, scientists) can find a different definition useful.

Here are some of them:

“GIS is much more than a container of maps in digital form”.

“A GIS is a computerized tool for solving geographic problems”

“GIS is a spatial decision support system”

“GIS is a mechanized inventory of geographically distributed features and facilities”

“GIS is a method for revealing patterns and processes in geographic information”

“GIS is a tool to automate time-consuming tasks that are too tedious or expensive or inaccurate if performed by hand”

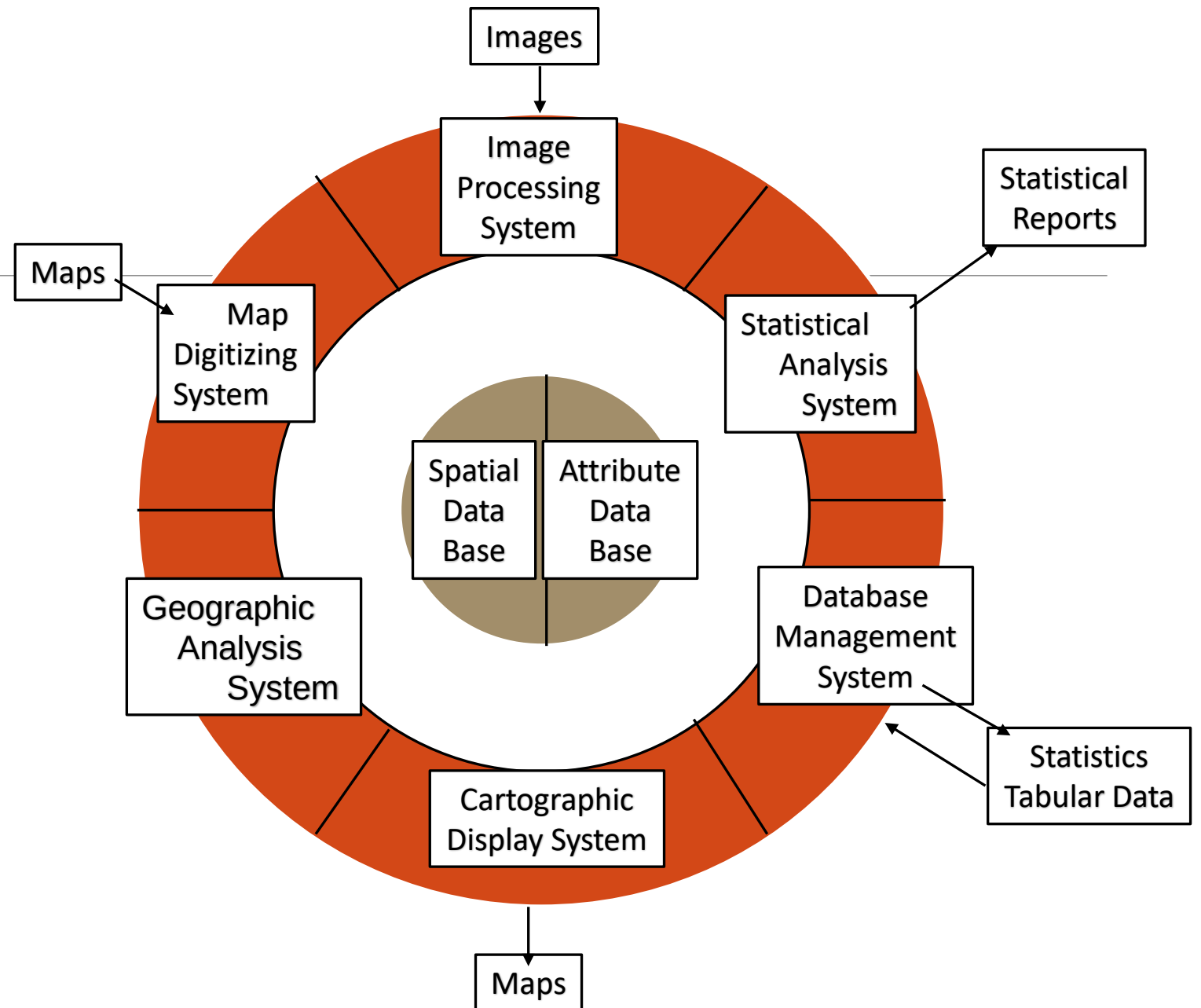
“GIS is a collection of computer hardware, software, and geographic data for capturing, managing, analyzing, and displaying all forms of geographically referenced information.”

What is GIS?

System used for

- Capturing
- Storing
- Updating
- Manipulating
- Analyzing and
- Presenting

Geographic Data



Is GIS about software *only*?

Information systems help us to manage *what we know*, by making easy to organise and store, access and retrieve, manipulate and synthesize, and apply knowledge to the solution of problems.

But is GIS about computers only?

Some definitions of GIS focus on the hardware, software, data and analysis of components. However, no GIS exist in isolation from the organizational context, and there must always be people to plan, implement and operate the system as well as make decision based on the output.

A Geographic Information System is not only about computers, software and electronic data.

A GIS is an organised collection of :

- Hardware
- Software
- Network
- Data
- Procedures
- **And people!**





Source: John Snow Inc. www.jsi.com



Dr. John Snow

and the 1854 Cholera outbreak in London's Broad Street region

Dr John Snow is known as the 'father of modern epidemiology' and the 'father of GIS' because of the famous case of the 1854 Cholera outbreak in London's Broad Street region.

In the 1850s, cholera was very poorly understood and massive outbreaks were a common occurrence in major industrial cities. An outbreak in London in 1854 in the Soho district was typical of the time, and the deaths it caused are shown in the map on the right.

Source: UCLA http://www.ph.ucla.edu/epi/snow/snowmap1_1854_lge.htm

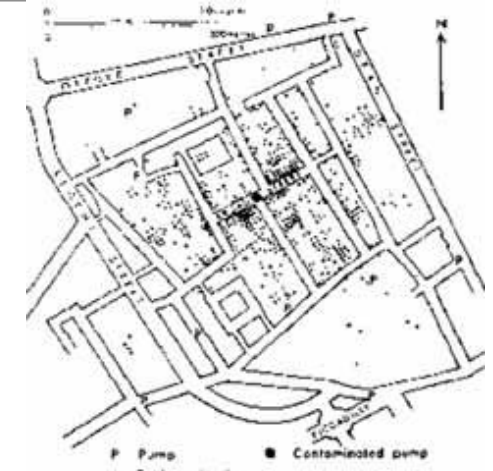
Text source: Longley et al (2005) *Geographic Information Systems and Science*. 2nd Edition. John Wiley and Sons Ltd. (Chapter 14, pages 317-319)

Dr John Snow

The map was made by Dr John Snow, who has conceived the hypothesis that cholera was transmitted through the drinking of polluted water, rather than through the air, as was commonly believed. He noticed that the outbreak appeared to be centred on a public drinking water pump in Broad Street – and if this hypothesis was correct, the pattern shown on the map would reflect the locations of people who drank the pump's water. There were anomalies, in the sense that deaths had occurred in households that were located closer to other sources of water, but he was able to confirm that these households also drew their water from the Broad Street pump. Snow had **the handle of the pump removed**, and the outbreak subsided, providing direct causal evidence in favour of his hypothesis.



Source:
http://blogs.esri.com/Info/photos/geography_matters/images/1440/secondarythumb.aspx



Source:
<http://www.gisdevelopment.net/application/health/overview/images/image1.jpg>

Dr. John Snow

It is worth investigating a bit more about this case and, if you live in London or plan to make a trip to the city, I suggest you visit the site where the story actually took place. There is a pub called 'John Snow Pub' located in Broadwick street where you can find a lot about the story. You can also see the place where the pump was originally located as well as a replica of the pump.



The pump with no handle is a replica of the famous Broad Street pump.

Source: <http://www.ph.ucla.edu/epi/Snow/snowpub.html>

The original pump was situated within a long-step on Broadwick Street of what is now the back wall of the John Snow pub. The location is currently marked with a curbstone and a capped water pipe.



Source: <http://www.ph.ucla.edu/epi/Snow/snowpub.html>

Dr John Snow

This Dr John Snow **story is about spatial analysis** or, in other words, on how the location in which events take place can have help us understand better the nature of phenomenon.

At the time of the breakout, cholera was not understood to be geographically related. However, as soon as location of deaths was introduced into the picture it was clear that this was a geographical problem.

The most interesting point of this story is that there were no computers used for the analysis. But we still consider it to be a story about GIS. Why is that?

More information about John Snow can be found at:

Brody H et al 2000 'Map-making and myth making in Broad Street: the London cholera epidemic, 1854'. The Lancet 356(9223) 64–8

Johnson, Stevenson (2006) Saviour of the city. The Times, 29 November 2006. Available online at:

http://www.timesonline.co.uk/article/0,,682-2475618_3,00.html

Johnson, Stevenson (2006) "The Ghost map". London: Penguin.

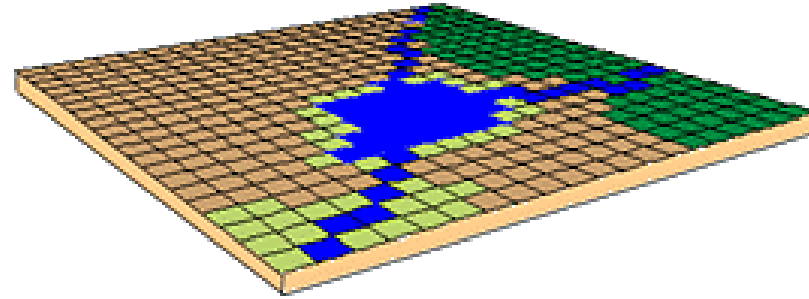
www.jsi.com

www.ph.ucla.edu/epi/snow.html

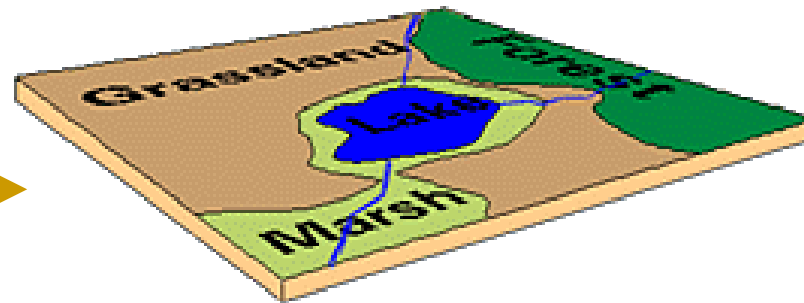
http://www.edwardtufte.com/bboard/q-and-a-fetch-msg?msg_id=0002Je&topic_id=1&topic=Ask+E%2eT%2e

Representing Spatial Elements

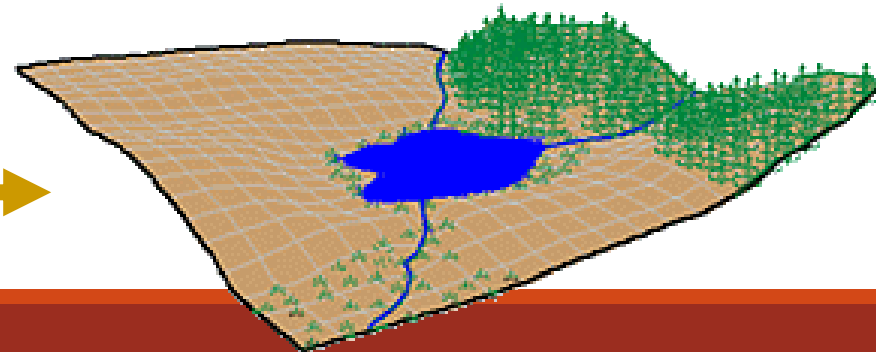
- RASTER



- VECTOR



- Real World



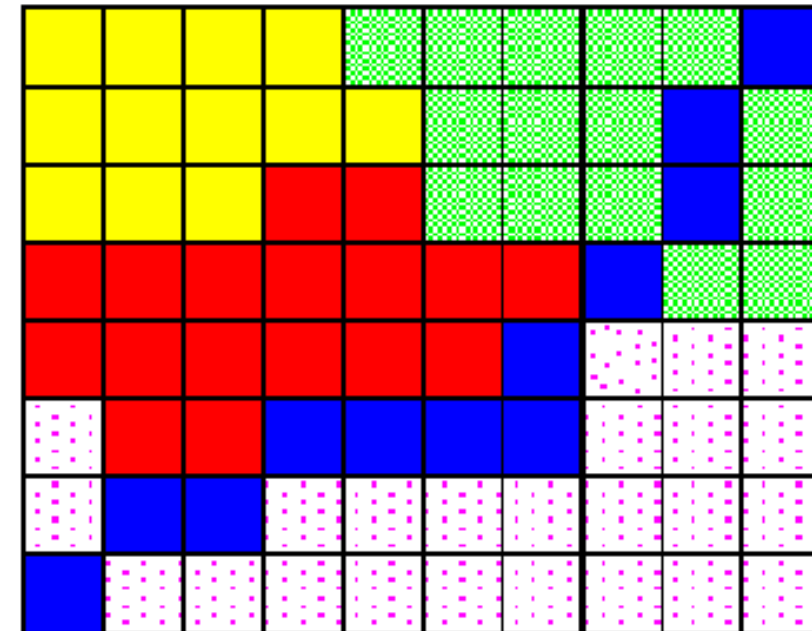
Representing Spatial Elements - **Raster**

Stores images as rows and columns of numbers with a Digital Value/Number (DN) for each cell.

Units are usually represented as square grid cells that are uniform in size.

Data is classified as “***continuous***” (such as in an image), or “***thematic***” (where each cell denotes a feature type).

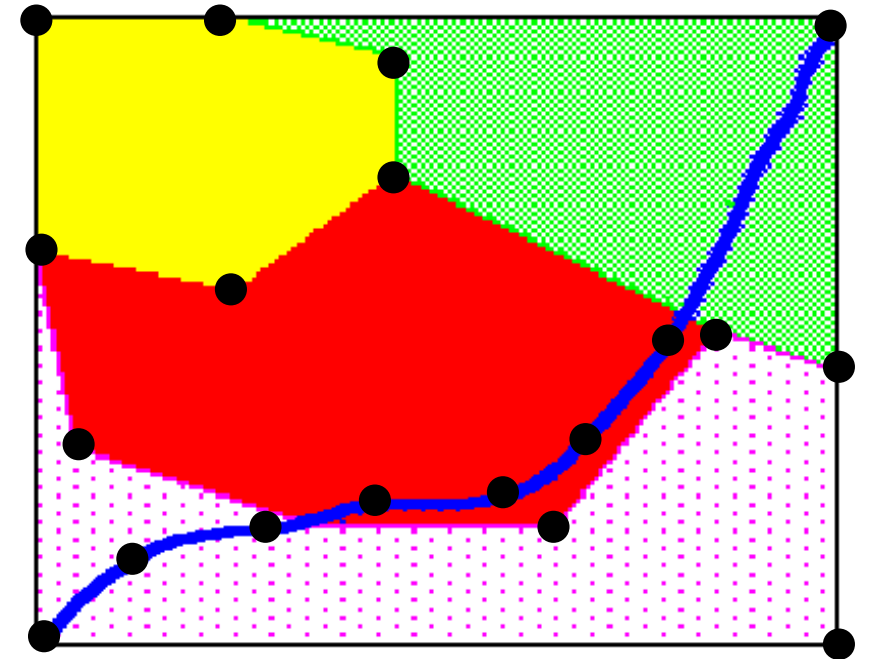
Numerous data formats (TIFF, GIF, ERDAS.img etc)



Representing Spatial Elements - **Vector**

Allows user to specify specific spatial locations and assumes that geographic space is continuous, not broken up into discrete grid squares

We store features as sets of X,Y coordinate pairs.



Representing Spatial Elements - Attributes

In the raster data model, the cell value (Digital Number) is the attribute. Examples: brightness, land cover code, Cost, Distance etc.

For vector data, attribute records are linked to point, line & polygon features. Can store *multiple* attributes per feature. Vector features are linked to attributes by a *unique feature number*.

Raster vs Vector

Raster Advantages

- The most common data format
- Easy to perform mathematical and overlay operations
- Satellite information is easily incorporated
- Better represents “continuous”- type data

Vector Advantages

- Accurate positional information that is best for storing discrete thematic features (e.g., roads, shorelines, sea-bed features).
- Compact data storage requirements
- Can associate unlimited numbers of attributes with specific features

Types of GIS

There are a number of Geographical Information Systems (GIS) (or GIS software) available today. They range from high-powered analytical software to visual web applications, and each of those are used for a different purpose.

Web-based GIS: ONS and London Profiler

Geobrowser: Google Earth, Bhuvan, Wikimapia

Desktop GIS: ArcGIS, QGIS, Grass

Web-based GIS

Web-based GIS, or WebGIS, are online GIS applications which in most cases are excellent data visualisation tools. Their **functionality is limited** compared to software stored on your computer, but they are user-friendly and particularly useful as they not required data download.

Geobrowser

A Geobrowser is better explained with reference to an internet browser, i.e. Internet Explorer. In short, a geobrowser can be understood as an Internet Explorer for geographic information. Like the internet it allows the combination of many types of geographic data from many different sources. The biggest difference between the World Wide Web and the geographic web however is that everything within the latter is *spatially referenced*.

Google Earth is the most popular geobrowser available and will be the one used for this course.

Desktop GIS

A GIS, or GIS software, allows you to interactively work with spatial data. A desktop GIS is a mapping software that needs to be installed onto and runs on a personal computer.

ArcGIS

QGIS

Grass

IDRISI

Data Sources

Bhuvan http://bhuvan.nrsc.gov.in/bhuvan_links.php#

USGS <http://earthexplorer.usgs.gov/>

Open Street Maps <https://www.openstreetmap.org/history#map=7/11.216/78.047>

ASTER <https://asterweb.jpl.nasa.gov/>

Q & A

If you want to propose a solution for water logging problem in your RWA

1. What all data you need?
2. Where to get it?
3. How do you represent each type of data?
4. How do you analyse?
5. How do you represent solutions?
6. How do you disseminate solutions to public for their opinion?